Predictive Modeling Analytic Plan

DSE6111

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To address ABC Corporation's desire to identify customers with a high risk of attrition, I propose designing, training, and testing a supervised classification learning model using customer data provided by ABC Corp. In this model, I expect to predict a customer's risk of attrition, using a response called **Attribution_Risk**. ABC Corporation will be able to identify at-risk customers and apply specific marketing tactics to prevent attrition. Additionally, I will provide a list of attributes that highly influence attrition risk prediction.

The customer data has been preliminarily reviewed for missing or invalid values, and none has been identified. I have noted a few values that need some cleansing, but these are very limited. All the data provided appears to be useful for this purpose. I am considering coding some categorical variables to numeric values. Candidate variables include:

- Gender
- Education_Level
- Marital_Status
- Income_Category
- Card_Category

Initially, I expect to use these quantitative features:

- Months_on_book
- Total_Relationship_Count
- Months_Inactive_12_mon
- Contacts_Count_12_mon
- Total_Amt_Chng_Q4_Q1
- Total_Trans_Amt
- Total_Trans_Ct
- Total_Ct_Chng_Q4_Q1
- Avg_Utilization_Ratio

I expect to use all features as the model matures.